Build and Run a Java Application with Maven and Migrate the Same Application to Grade

Step 1 : Build and Run a Java Application with Maven

To build a basic Java Application with Maven Project, run following command:
**mvn archetype: generate -DgroupId=com.example -DartifactId=maven-to-gradle \
-DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -Da
rchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 23 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shravan-hegde/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/Devops/MyMavenToGradle
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  1.765 s
[INFO] Finished at: 2026-03-25T08:57:12+05:30
[INFO] ----------------------------------------------------------------------------
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ cd MyMavenToGradle/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$
```

Step 2 : Go to MyMavenToGradle directory using cd command and use tree command to see the file structure

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ tree
.
├── pom.xml
└── src
    ├── main
    │   └── java
    │       └── com
    │           └── example
    │               └── App.java
    └── test
        └── java
            └── com
                └── example
                    └── AppTest.java

10 directories, 3 files
```

Step 3 : Compile and Install Maven Project

Run the mvn clean install command to compile the Java source code, execute unit tests, and package the application into a JAR file located in the target directory

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MyMavenToGradle >---------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenToGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenToGradle/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/shravan-hegde/Devops/MyMavenToGradle/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/shravan-hegde/Devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenToGradle ---
[INFO]
```

Step 4 : Verify Successful Maven Build and Execution

Confirm the build is successful and execute the generated JAR file using the java -jar command to ensure the "Hello World!" application runs correctly before migration.

```
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenToGradle ---
[INFO] Building jar: /home/shravan-hegde/Devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenToGradle ---
[INFO] Installing /home/shravan-hegde/Devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar to /home/shravan-hegde/.m2/reposito
ry/com/example/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO] Installing /home/shravan-hegde/Devops/MyMavenToGradle/pom.xml to /home/shravan-hegde/.m2/repository/com/example/MyMavenToGradle/1
.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.968 s
[INFO] Finished at: 2026-03-25T09:06:04+05:30
[INFO] ------------------------------------------------------------------------
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ java -jar target/MyMavenToGradle-1.0-SNAPSHOT.jar
Hello World!
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$
```

Step 5 : Troubleshoot Migration Command

Identify the correct migration syntax by attempting the gradle init command, noting that Gradle requires specific supported types such as pom to initiate the conversion.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ gradle init --type pom.xml
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported types: 'basic', 'groovy-application', 'groovy-library', 'java-app
lication', 'java-library', 'pom', 'scala-library'.

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log output. Run with --scan to get fu
ll insights.

* Get more help at https://help.gradle.org

BUILD FAILED in 0s
2 actionable tasks: 2 executed
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$
```

Step 6 : Convert Maven Project to Gradle

Execute the gradle init --type pom command to trigger the "incubating" migration feature, which automatically parses the pom.xml to generate equivalent Gradle build scripts.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ gradle init --type pom
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$
```

Step 7 : Verify Migrated Gradle Files

Run the ls command to confirm that Gradle has successfully added new configuration files like build.gradle, settings.gradle, and the gradlew wrapper alongside the original Maven files.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$ ls
build.gradle  gradle  gradlew  gradlew.bat  pom.xml  settings.gradle  src  target
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenToGradle$
```